

DRAFT — DBA-OIL-FC2: Offshore Crude Production

Offshore Crude Production

Design Basis Accident — Military Action

DBA-OIL-FC2 — Facility Class Document

Issued Under: DBA-OIL Sector Framework

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Preface

The offshore crude production platform is unique among petroleum sector facility classes. It is a major industrial installation — capable of producing tens of thousands of barrels of crude oil per day — situated in a remote maritime environment far from the emergency response resources, spare parts logistics, and specialist contractor capacity available to onshore facilities. When a platform is damaged, recovery does not begin when the fire is out. It begins when the weather permits access, when specialized offshore construction vessels can be contracted and mobilized, and when the regulatory inspection process authorizes restart. These constraints are not bureaucratic inconveniences — they are physical realities that an adversary understands and exploits.

This document establishes the Design Basis Accident scenarios for deliberate military action against offshore crude production infrastructure. The primary reference environment is the Gulf of Mexico outer continental shelf, where approximately fifteen percent of total U.S. crude production is concentrated on a relatively small number of major platforms and deepwater installations. The allied-nation reference environments are the North Sea and the Norwegian Continental Shelf, where European energy security depends on offshore production infrastructure that faces an analogous maritime threat profile.

The evidentiary anchors for this document are three. The 2019 Abqaiq-Khuras attack demonstrated that cruise missile and drone strikes against major energy production infrastructure can produce immediate, large-scale supply disruption; the offshore adaptation of that template eliminates the land-based access advantages that enabled the rapid Saudi Aramco recovery. The 2022 Nord Stream pipeline sabotage demonstrated that state-actor subsea operations against energy infrastructure are operationally feasible, attributionally deniable, and consequence-generating at a scale that affects alliance energy security. The Deepwater Horizon blowout (2010) — while non-adversarial — established the governing reference for the regulatory consequence cascade following a major offshore event and defines the non-adversarial bounding case against which the military-action extension is built.

The lessons from these three events converge on a single analytical conclusion: the offshore production platform is a high-consequence, low-resilience target whose remote operating environment transforms a successful attack into an extended supply disruption that no available onshore response mechanism can quickly offset.

1. Introduction

1.1 Purpose

This document establishes the Design Basis Accident — Military Action scenarios for the offshore crude production facility class. Its purpose is to define the bounding threat event that offshore production security planning, resilience investments, and recovery architecture must

address. As with all DBA-MA documents, this document does not prescribe specific protective measures at individual facilities. It defines the consequence envelope within which protective measures must perform.

An offshore production security program designed against a surface-threat model — physical security of the platform against boarding or small-boat attack — will fail when a sophisticated adversary integrates cyber pre-positioning, precision standoff kinetic action, and subsea sabotage in a coordinated sequence. The DBA-MA approach defines the coordinated, multi-vector, multi-domain attack as the governing design event.

1.2 Scope and Applicability

This document addresses offshore crude production facilities: platforms and associated subsea and export pipeline infrastructure located on the outer continental shelf and in deepwater operating environments. The facility class encompasses fixed platforms, tension leg platforms (TLPs), SPAR platforms, floating production storage and offloading vessels (FPSOs), and their associated subsea wellheads, flow lines, risers, and export pipelines to the point of landfall.

Onshore crude oil production, petroleum refining, and product pipeline systems are addressed in their respective DBA-OIL facility-class documents. The export pipeline is within the scope of this document from the platform to the point of landfall; the onshore trunk pipeline system from landfall to the refinery is within the scope of DBA-OIL-FC4. Where events affecting the export pipeline propagate consequences onshore, this document addresses those consequences as they originate at the platform without displacing the FC4 analysis of the onshore pipeline system.

Natural gas produced in association with crude oil at offshore platforms — associated gas — is addressed here insofar as its loss is a consequence of the crude production disruption. Dedicated offshore gas production facilities are addressed in the DBA-GAS series.

1.3 Relationship to Governing Documents

This document is issued under the DBA-OIL Sector Framework (v1.7), and is governed by DBA-EN-SF: Energy Sector Design Basis Framework (v0.8), which is itself issued under the DBA-MA Parent Framework (IAEA L1 v1.4). The threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards are inherited from the sector framework. This facility-class document adds the offshore-production-specific physical configuration, asset taxonomy, bounding events, and consequence chains that the sector framework establishes in general terms.

The inheritance map for this document is as follows. The threat actor taxonomy (Tier 1, 2, 3), the four characteristics of military action (intentionality, persistence, precision, denial), the three-tier consequence framework, and the adequacy standards are inherited unchanged from the sector framework. The bounding event categories — process safety events, infrastructure attack events, natural hazard events, and operational events — are inherited with offshore-specific adaptation.

The Dual Design Basis architecture and Bright Line / Command Broker requirements are inherited unchanged and applied to the offshore control system environment in Section 6.

1.4 The Offshore Platform as a Remote Single-Node Asset

A fundamental characteristic of the offshore production facility class distinguishes it from every other facility class in the DBA-OIL series: the offshore platform combines high production consequence with extreme access restriction. A refinery attack's consequences are governed by component lead times and specialty contractor availability. An offshore platform attack's consequences are governed first by the maritime environment itself — weather windows, sea state, water depth, and distance from shore — and only secondarily by component procurement.

This distinction drives the entire DBA-MA analysis. The four characteristics of military action apply with particular force to the offshore environment: intentionality allows target selection of the highest-production platform in the cluster; persistence means pre-positioned cyber intrusion can wait for the optimal weather and operational window; precision allows standoff kinetic action against the highest-consequence topside equipment without requiring access to the platform; and denial — the most powerful characteristic in the offshore context — is partly provided by the environment itself. A state actor attacking an offshore platform does not need to actively deny access to the response. The Gulf of Mexico does that through weather, distance, and water depth.

2. Offshore Production Profile and Asset Taxonomy

2.1 Platform Types and Physical Configuration

Fixed Platforms. Jacket-type structures on the seabed, connected to surface topsides by steel tubular legs. Fixed platforms are limited to water depths generally below 1,500 feet and are concentrated in the shallow-water Gulf of Mexico shelf. They are the most numerous platform type in U.S. offshore production, but individually account for a smaller share of total production than deepwater installations. Fixed platforms are physically more accessible for both maintenance and attack than deepwater floating platforms.

Tension Leg Platforms (TLPs). Floating platforms are moored to the seabed by tensioned tendons. TLPs are used in water depths from approximately 1,000 to 5,000 feet. Several major Gulf of Mexico deepwater fields are produced from TLPs. The tendon system is a structural critical point: damage to one or more tendons can compromise platform stability and force evacuation regardless of topside damage.

SPAR Platforms. Deep-draft floating production systems consisting of a large cylindrical hull extending deep below the waterline, moored by catenary or semi-taut mooring lines. SPARs are deployed in water depths from approximately 2,000 to over 7,000 feet and account for a significant share of deepwater Gulf of Mexico production. Their deep draft provides stability in

high sea states and makes them less vulnerable to surface wave action than other platform types, but also means that subsea mooring line damage is a high-consequence attack vector.

Floating Production, Storage, and Offloading Vessels (FPSOs). Ship-shaped vessels that produce, process, and store crude oil, offloading periodically to shuttle tankers rather than through a fixed export pipeline. FPSOs are the dominant production system in the North Sea and Brazilian presalt fields and are increasingly deployed in the deepwater Gulf of Mexico. Their tanker-offloading architecture means that export pipeline sabotage does not interrupt production in the same way it does for pipeline-tied platforms — but the shuttle tanker becomes a target in its own right under military-action conditions.

2.2 Subsea Infrastructure

Modern deepwater fields use subsea production systems — wellheads, manifolds, flow lines, and risers — to connect producing wells on the seabed to the surface platform. Subsea infrastructure operates at extreme depths, in cold water, under high pressure, and without direct human access under normal operating conditions. Remotely operated vehicles (ROVs) are the primary tool for subsea inspection, maintenance, and intervention. Under military-action conditions, the ROV fleet becomes a critical resource. There are a limited number of deep-water-capable ROV support vessels globally, and their availability in a conflict environment where the adversary may interdict surface support operations constrains subsea recovery options.

The subsea export pipeline — or riser system connecting the platform to the export pipeline — is a consequence-amplifying target. Unlike topside equipment, which can sometimes be bypassed or temporarily substituted, a severed subsea export pipeline or damaged riser system stops all production from the affected platform regardless of topside integrity. The Nord Stream sabotage demonstrated the operational template at a pipeline scale; applied to a deepwater export riser or subsea pipeline segment, the same approach produces a platform-level production outage with a recovery timeline governed by ROV access, pipe-lay vessel availability, and water depth.

2.3 Gulf of Mexico Cluster Profile

The Gulf of Mexico outer continental shelf is the primary reference environment for DBA-OIL-FC2. Offshore GoM production accounts for approximately fifteen percent of total U.S. crude production, concentrated disproportionately in a relatively small number of major deepwater fields in the Mississippi Canyon, Green Canyon, Garden Banks, and Walker Ridge protraction areas. The top ten producing fields account for a substantial majority of total GoM production; a coordinated attack on the top three or four producing platforms would produce a supply reduction that the onshore producing regions cannot quickly compensate for, given that onshore capacity is already operating near its infrastructure limits.

GoM platforms are connected to the onshore pipeline system through a network of offshore gathering pipelines and trunk lines that converge at major hubs — the Poseidon, Mars,

Amberjack, and Hoover-Diana pipeline systems, among others — before making landfall at multiple points along the Louisiana and Texas coast. The hub architecture creates a second-order vulnerability: a pipeline hub attack can disrupt production from multiple platforms simultaneously, even if the platforms themselves are undamaged.

2.4 Allied-Nation Reference Environments

North Sea. The UK Continental Shelf and Norwegian Continental Shelf together produce approximately three to four million barrels per day of crude oil and condensate, a significant share of European energy supply. The Forties Pipeline System in the UK and the Statfjord, Troll, and Johan Sverdrup fields in Norway represent the high-consequence nodes in the allied offshore production system. The North Sea Transition Authority manages UK offshore production; the Norwegian Offshore Directorate manages Norwegian production under a regulatory framework that provides a useful comparison for Design Basis development. The North Sea operating environment — harsh weather, high sea states, and seasonal access limitations — amplifies the denial characteristic of military action in ways directly analogous to the GoM.

Norwegian Continental Shelf. Norway is the primary non-Russian supplier of natural gas to continental Europe and a significant crude oil exporter. The Troll, Oseberg, and Gullfaks fields are critical production nodes. Norwegian Petroleum Security legislation and the Norwegian Petroleum Directorate's security requirements provide the closest allied-nation analog to a formal offshore Design Basis framework, and Norwegian offshore security architecture should inform the adequacy standards developed in this and subsequent revisions of DBA-OIL-FC2.

2.5 Regulatory Authority

BSEE. The Bureau of Safety and Environmental Enforcement (30 CFR Parts 250 and 254) is the primary regulator for offshore oil and gas operations on the U.S. outer continental shelf. BSEE regulations govern well control, production safety systems, structural integrity, oil spill response planning, and safety management systems. The Well Control Rule (30 CFR Part 250, Subpart D) and the Production Safety Systems regulations (Subpart H) establish the baseline for the non-adversarial Design Basis. BSEE has no mandatory cybersecurity regulations for offshore platform control systems; this is the governing regulatory gap addressed in Section 7.

USCG. The United States Coast Guard (33 CFR) has jurisdiction over maritime security, marine environmental protection, and vessel safety for offshore facilities and associated supply vessels. USCG Maritime Security (MARSEC) levels trigger enhanced security measures at Outer Continental Shelf (OCS) facilities. The USCG's response role in a military-action scenario extends to maritime domain awareness, enforcement of exclusion zones, and coordination with the Navy for threat response.

PHMSA. The Pipeline and Hazardous Materials Safety Administration (49 CFR Part 195) regulates the offshore export pipeline from the production facility to the point at which it

becomes a hazardous liquid pipeline under onshore jurisdiction. The PHMSA offshore pipeline safety framework establishes integrity management requirements that provide the regulatory baseline for the export pipeline component of the Design Basis.

3. Design Basis: Non-Adversarial Bounding Events

3.1 DB Layer Purpose and Structure

The Design Basis layer establishes the cause-agnostic postulated events that an offshore crude production facility must be demonstrated to withstand, independent of whether the initiating condition is a process-safety or equipment failure, a natural hazard, an operational decision, or a cross-sector cascade. Version 0.1 §3.1 established three non-adversarial bounding events in narrative form. Version 0.2 formalizes those events into the series-standard event-code table structure established in DBA-OIL-FC4 v0.3, adds the operational event (OE) and cross-sector cascade (XC) categories omitted from v0.1, and separates the natural-hazard and subsea-infrastructure events into their own coded entries.

Four event categories organize the DB layer for the offshore crude production facility class: equipment, infrastructure, and process-safety failure (EF); natural hazard (NH); operational event (OE); and cross-sector cascade (XC), inheriting the process-safety, infrastructure-failure, natural-hazard, and operational categories of the DBA-OIL Sector Framework (§§4.3–4.5). Two properties of the offshore facility class govern throughout: it is a remote single-node asset whose repair depends on scarce marine intervention assets (the offshore construction vessel, §4.5); and it sits at the head of the wellhead → export pipeline → refinery chain, so its loss propagates downstream as a crude-supply consequence measured by SDD.

3.2 Equipment, Infrastructure, and Process-Safety Failure (EF) Postulated Events

Code	Event	Initiating Condition and Governing Concern
EF-1	Platform fire and explosion / loss of well control	An uncontrolled topside hydrocarbon release escalating to fire and explosion, or a loss of well control / blowout. Piper Alpha (1988) is the reference process-safety event — a seal failure cascaded through a congested topside to a gas-riser rupture, destroying the platform and killing 167. Deepwater Horizon (2010) is the reference well-control event, adding an 87-day uncontrolled subsea release and a six-month basin drilling moratorium. Governing concern: total or extended platform production loss, with the blowout case carrying a regulatory extended-consequence regime beyond physical reconstruction.
EF-2	Subsea infrastructure failure	Loss of well control or integrity failure of a subsea component — flowline, riser, manifold, umbilical — at a deepwater installation, removing production from all wells tied back to the affected subsea system until the component can be accessed and repaired by ROV or diver intervention. At extreme water depth, intervention may require days to weeks and is

		governed by weather and marine-vessel availability (\$4.5).
EF-3	Export pipeline / onshore receiving failure	Failure of the subsea export pipeline or the onshore receiving terminal that strands otherwise-intact platform production by removing its delivery path. Manifests the wellhead → export pipeline → refinery dependency: production capacity exists but cannot reach the served refining system.

3.3 Natural Hazard (NH) Postulated Events

Code	Event	Initiating Condition and Governing Concern
NH-1	Hurricane / tropical cyclone (Gulf of Mexico)	A major hurricane traversing the producing basin, forcing evacuation and shut-in of essentially all exposed production. Ivan (2004) and Katrina/Rita (2005) are the reference events — hundreds of platforms damaged and GoM output reduced by roughly one million barrels per day for months. Bounding case: a direct platform strike compounded with a pipeline-hub disruption, requiring both platform repair and pipeline restoration before the affected cluster resumes. The dominant offshore natural hazard.
NH-2	Loop Current and extreme metocean conditions	Strong loop-current and severe sea-state conditions degrading deepwater station-keeping (dynamic positioning and mooring), inducing riser fatigue and forcing production curtailment or disconnection at floating facilities.
NH-3	Subsea geohazard / seismic event	Seafloor slope failure, mudslide, or seismic motion affecting subsea pipelines and infrastructure — a recognized Gulf of Mexico mudslide exposure for pipeline integrity — producing subsea-component loss with the EF-2 intervention-constraint consequence.

3.4 Operational Event (OE) Postulated Events

Code	Event	Initiating Condition and Governing Concern
OE-1	Production allocation or export-scheduling error	Operational mismanagement of platform or field throughput, or of export nominations, reducing delivered crude without a physical failure. The non-adversarial baseline for delivery-shortfall conditions; the adequacy standard is allocation review against independent production measurement.
OE-2	Deferred subsea intervention / maintenance-availability shortfall	Deferred inspection or maintenance of subsea components, or an extended planned turnaround, that creates an availability gap and lengthens the response to a subsequent failure. Establishes, at the DB layer, the vessel-and-intervention availability margin that the Scenario C cluster attack deliberately exhausts.

3.5 Cross-Sector Cascade (XC) Postulated Events

Code	Event / Source	FC2 Consequence
XC-1	Electric / grid dependency at onshore receiving and export control (DBA-ES source event)	Onshore receiving terminals, metering, and export control depend on grid power; a DBA-ES event degrades the export path even while platforms produce. The electricity-to-oil direction of the cross-sector dependency.

XC-2	Offshore crude loss → refining feedstock cascade (DBA-OIL-FC1)	Loss of offshore crude reduces refinery feedstock, cascading to refined-product supply through the wellhead → export → refinery → product chain — the oil-internal cascade, measured by SDD at the served refining system.
XC-3	Associated-gas cross-link to the gas sector (DBA-GAS interface)	Offshore oil production co-produces associated gas feeding gas processing and pipelines; platform loss removes that associated-gas supply — the oil-to-gas direction of the bidirectional energy interface.

4. Design Basis Accident: Non-Adversarial Consequence Envelope

4.1 Bounding Event Identification

The DBA layer identifies the bounding events from the DB layer that define the consequence envelope within which protective measures and recovery architecture must perform. Three bounding events govern the offshore crude production facility class.

Bounding Event	Source	Governing Consequence	Metric
Platform fire/explosion or blowout	EF-1	Total or extended loss of platform production; for the blowout case, an additional regulatory moratorium across the basin beyond physical reconstruction.	SDD; platform reconstruction duration (+ regulatory hold)
Subsea hub failure — multi-platform tieback loss	EF-2	Loss of all production tied back to the affected subsea system until marine intervention; governed by offshore construction vessel availability, not component lead time alone.	SDD; vessel-bound intervention duration
Hurricane platform-and-hub compound	NH-1	Basin-scale evacuation and shut-in with combined platform and pipeline-hub damage requiring dual restoration before the cluster resumes.	SDD; basin restoration duration

4.2 SDD — Supply Delivery Duration Metric Definition

The offshore crude production facility class is governed by Supply Delivery Duration (SDD), the oil-sector series-standard metric established in DBA-OIL-FC4 and sanctioned at the DBA-EN-SF level. For the offshore class, SDD measures the continuous interval during which lost offshore production prevents the served refining and export system from receiving its required crude delivery. SDD is measured at the onshore delivery point of the wellhead → export pipeline → refinery chain, not at the platform, so it captures both production loss and export-path loss (EF-3) on a common basis.

4.3 Non-Adversarial Consequence Taxonomy

Tier 1 — Facility/cluster-level. Loss of a platform or a subsea-hub cluster's production for the governing recovery duration, contained within the affected installation or tieback.

Tier 2 — Basin/regional-level. Reduction of basin crude output sufficient to force draws on inventory and alternative supply and to constrain the served refining system, measured by SDD at the onshore delivery point.

Tier 3 — Cross-sector cascade. Crude-supply loss propagating to refining feedstock (XC-2 → refined product shortfall), to associated-gas supply (XC-3), and, for allied-nation reference environments, to strategic and coalition supply dependencies.

4.4 SDD Acceptance Criterion Framework

SDD follows the series Zone 1/2/3 acceptance-criterion framework with T1/T2 calibration thresholds; site-specific calibration is required at implementation. T1 (inventory-and-alternative-supply bridge) is the interval over which crude inventory, strategic reserves, and alternative-supply ramp can offset the lost offshore production before the served refining system is constrained. T2 (intervention/reconstruction bridge) is the interval governed by marine-intervention and reconstruction duration — the offshore construction vessel constraint (§4.5). SDD is not additive across platforms; where multiple installations in a basin are affected, the governing SDD is the longest single critical path against the regional delivery requirement, not the cumulative sum.

4.5 RT_HVLLC Application — Offshore Construction Vessel as Governing Long-Lead Constraint

For the offshore facility class, the governing long-lead constraint is not a single replaceable component but the offshore construction vessel — heavy-lift, derrick, pipelay, and dynamically positioned construction and intervention vessels. This is the offshore analog of the pipeline pump-unit strategic-spare gap identified for DBA-OIL-FC4: the global fleet of such vessels is small and booked far ahead, so subsea and platform repair is availability-bound rather than component-bound. RT_HVLLC for the offshore class is therefore expressed as vessel-mobilization-plus-intervention duration, and is the binding term in T2. The MA extension (§5) bounds the case in which the adversary attacks multiple installations simultaneously to overwhelm the vessel pool — the Scenario C mechanism.

5. DBA-MA Military Action Extension

5.1 MA Extension Foundation

The DBA-MA extension applies the four MA characteristics — intentionality, persistence, precision, and denial — to the non-adversarial bounding events established in §§3 and 4. The three scenarios are carried forward from v0.1 without change to their analytical content; v0.2 grounds each in the DB/DBA events it exploits. Scenario A (coordinated cyber-kinetic platform attack) is the EF-1 platform fire/explosion mechanism deliberately induced. Scenario B (subsea pipeline hub sabotage) is the EF-2 subsea-infrastructure failure directed at a shared hub to maximize multi-platform tieback loss. Scenario C (multi-platform cluster attack) is the compound that deliberately exhausts the offshore construction vessel pool (§4.5), converting a recoverable production loss into an extended basin-supply consequence. The remoteness and maritime threat profile of the offshore environment are themselves instruments of denial against intervention and repair.

Pending dependency. This instrument does not yet carry the four-filter MA applicability assessment or the CB-1 through CB-5 MA communications design basis. Both frameworks are currently electric-sector-scoped (DBA-ES-MA-SF / DBA-ES-MA-QB) with no oil-sector anchor; they are deferred pending a DBA-OIL sector-level MA companion (proposed DBA-OIL-MA-SF), and are flagged here at the point where an operator first engages the MA extension.

The DBA-MA bounding event for the offshore crude production facility class is a coordinated, multi-vector, multi-domain attack against a major deepwater production platform or a cluster of major platforms in the Gulf of Mexico, executed during a high-production period and designed to exploit the remote operating environment as a force multiplier for the denial characteristic.

The bounding parameters, inherited from the sector framework threat taxonomy:

Adversary tier: Tier 1 state actor with maritime special operations capability, advanced cyber pre-positioning, and precision standoff strike capability.

Modality: Multi-domain combined arms. Subsea sabotage of the export pipeline or riser system to deny production export; cyber compromise of the platform control system to manipulate production conditions and suppress safety system response; and precision standoff kinetic action against the highest-consequence topside equipment — gas compression, produced water handling, or the export riser system — timed to exploit the control system compromise window.

Target selection: The highest-producing platform or platforms in the Gulf of Mexico cluster, with secondary targeting of the pipeline hub through which their production flows. The adversary selects targets to maximize the compound effect: platform damage that cannot be repaired until the export pipeline is restored, and pipeline damage that cannot be addressed until the platform is safe to access.

Timing: A weather window that permits the subsea and kinetic components to execute while simultaneously restricting the response — heavy weather that grounds helicopter operations and limits surface vessel access following the attack. The adversary uses the GoM weather system as an operational asset.

Simultaneity: Attacks on two or more major platforms or pipeline hubs within a window that exhausts the Gulf of Mexico's specialized offshore construction vessel capacity — the offshore analog of the specialty contractor constraint identified in the FC1 cluster attack scenario.

5.2 Four MA Characteristics Applied to Offshore Crude Production

MA Characteristic	Application to Offshore Crude Production	Governing Concern
Intentionality	The adversary selects the platform or shared subsea hub whose loss removes the most basin production and most directly forces the offshore-construction-vessel bottleneck; allied-nation reference environments extend the target set to coalition supply.	A shared subsea hub is a single high-consequence node; one precise action removes many platforms' production.
Persistence	Subsea and platform repair is inherently vessel-availability-bound and measured in months; persistence is the reconstruction timeline itself, extended by denial of vessel and crew access, and — for the blowout case — by a regulatory moratorium.	Persistence converts a recoverable production loss into an extended basin-supply emergency with no further action.
Precision	Precision selects the specific subsea manifold, riser, or topside process-safety system whose loss yields maximum tieback or escalation consequence with a minimal, low-signature footprint.	Precision against a process-safety system can induce the EF-1 escalation; precision against a hub maximizes the vessel burden.
Denial	Denial operates through the remoteness of the offshore environment and a sustained maritime threat that denies ROV, diver, vessel, and crew access during intervention.	Denial of marine intervention access is the offshore class's governing MA vulnerability — it removes the recovery window directly.

5.3 Scenario A: Coordinated Cyber-Kinetic Platform Attack

5.3.1 Initiating Event

Simultaneous cyber-kinetic attack against a major deepwater SPAR or TLP platform in the Gulf of Mexico, producing approximately 80,000 to 120,000 barrels per day. The cyber component, having achieved process control network access through a multi-stage intrusion campaign exploiting the platform's satellite communications link over the preceding months, initiates coordinated manipulation of the gas compression system and the produced water handling system. The manipulation is designed to create an overpressure condition in the gas export riser while simultaneously causing a produced water pump seal failure — the combination seeding a hydrocarbon release at two locations on the production deck.

The DCS manipulation suppresses the relevant high-pressure alarms and inhibits the ESD actuation for a defined time window. The kinetic component executes within that window: precision munitions or UAS-delivered charges against the gas compression module and the export riser PLEM (pipeline end manifold) at the seabed. The platform fire and the subsea export pipeline breach occur simultaneously.

5.3.2 Progression Sequence

T+0: Gas compression overpressure event and produced water seal failure produce hydrocarbon releases on the production deck. Partial alarm suppression prevents immediate detection. Kinetic strikes against the gas compression module and the subsea PLEM execute. Platform ESD is partially actuated; inhibition prevents full automatic shutdown of the affected wells.

T+0 to T+2 hours: Fire on production deck escalating as well, flow continues through compromised ESD. Crew evacuation initiated. Helicopter operations are hampered by fire and smoke. The platform control room is operating from compromised DCS data. Subsea PLEM breach confirmed by production drop — all production from the platform ceases as the export pipeline depressurizes.

T+2 to T+24 hours: Full crew evacuation complete. Platform fire continues; production deck structure sustaining thermal damage. Coast Guard and support vessel response delayed by weather and distance. Subsea PLEM breach producing ongoing hydrocarbon release to the GoM. BSEE notification and Coast Guard incident command activated. Well shut-in is being attempted through subsea wellhead closure — complicated by DCS compromise that has propagated to the subsea control system.

T+24 to T+72 hours: Platform fire suppressed by burnout or limited firefighting response. Damage assessment cannot begin until the platform is safe for ROV survey and personnel boarding under improved weather. Subsea PLEM breach confirmed by ROV inspection: the pipeline end manifold requires replacement, not repair. Pipe-lay vessel procurement initiated. Wells shut in at the subsea wellhead; production loss of 80,000 to 120,000 barrels per day confirmed for the duration of recovery.

T+72 hours onward: DCS forensic investigation ongoing at the onshore support center. Physical damage assessment proceeding by ROV. Recovery timeline estimate: pipe-lay vessel mobilization 4 to 8 weeks; PLEM procurement 8 to 16 weeks; topside repair assessment pending structural survey; regulatory inspection and restart authorization timeline to be determined following BSEE review.

5.3.3 Consequence Envelope

Immediate production impact. Total loss of platform production — 80,000 to 120,000 barrels per day — from the moment of the export pipeline breach. Unlike a topside-only event, where a partial production restart might be possible through tanker offloading, the pipeline-tied platform

has no alternative export path. Production loss is complete and continues until the PLEM is replaced and the export pipeline is restored.

Recovery timeline. The governing constraint is the pipe-lay vessel schedule. Heavy-lift pipe-lay vessels capable of operating at deepwater GoM depths are a global finite resource; there are approximately a dozen vessels worldwide capable of deep-water pipeline work. In a scenario where multiple GoM platforms or pipelines have been attacked simultaneously (Scenario C), competition for pipe-lay vessel capacity extends each facility's recovery timeline. The PLEM replacement timeline of 8 to 16 weeks, combined with topside repair and regulatory restart authorization, produces a total production outage of 4 to 9 months for the primary consequence envelope.

Environmental consequence. The subsea PLEM breach produces an ongoing hydrocarbon release until the wells are shut in at the subsea wellhead. Depending on the time to achieve subsea well shut-in — complicated by the DCS compromise — the release volume may be significant. The regulatory consequence cascade from a major GoM hydrocarbon release — BSEE investigation, potential drilling and production moratorium in the affected area — is an embedded component of the consequence envelope, as demonstrated by the Deepwater Horizon precedent.

5.4 Scenario B: Subsea Pipeline Hub Sabotage

5.4.1 Initiating Event

Subsea sabotage of a major Gulf of Mexico pipeline hub — a gathering system junction through which production from multiple platforms flows before being transported to shore. The attack employs the Nord Stream operational template: swimmer delivery vehicles or autonomous undersea vehicles delivering shaped charges against the hub manifold or the pipeline at a point where seabed terrain or pipeline geometry concentrates stress. The attack is executed without warning during normal platform operations, producing a sudden, complete loss of export capability for all platforms connected to the hub.

5.4.2 Multi-Platform Production Loss

The pipeline hub architecture of the Gulf of Mexico — where a relatively small number of gathering hubs aggregate production from multiple platforms before conveying it to shore — creates an attack surface that multiplies the consequence of a single successful subsea operation. A hub that aggregates 200,000 to 400,000 barrels per day of production from four to eight platforms represents a target whose successful attack produces a consequence two to four times greater than a single-platform attack, using an operationally simpler attack vector.

The platforms connected to the damaged hub have no immediate alternative export path. They shut in production at the wellhead and wait. The platform crews may be evacuated if the hub

damage creates a safety concern for the connected riser systems; the platforms themselves are physically undamaged but completely non-productive.

5.4.3 Consequence Envelope: Hub Attack

Supply impact. 200,000 to 400,000 barrels per day of GoM crude production is offline for the duration of hub repair. The hub repair requires a pipe-lay vessel and, depending on the extent of damage, replacement of the manifold structure. The repair timeline parallels the PLEM replacement timeline in Scenario A: 4 to 8 weeks for vessel mobilization, 8 to 20 weeks for the repair itself, plus regulatory inspection and restart.

Attribution and response. The subsea sabotage template is designed for deniability. Unlike a kinetic strike, which is immediately visible and attributable, a subsea pipeline failure requires investigation to distinguish sabotage from a corrosion or anchor-drag failure. Attribution may take weeks. The adversary uses the attribution delay to constrain the political response: a nation that cannot quickly prove sabotage cannot quickly invoke alliance response mechanisms.

Allied analog. The Nord Stream sabotage produced a permanent loss of pipeline capacity — the pipelines were never repaired — because the political and economic context rendered repair non-viable. The GoM analog would not produce a permanent loss. Still, the deniability of the attack and the extended repair timeline produce a sustained supply disruption that achieves the adversary's strategic objective regardless of eventual attribution.

5.5 Scenario C: Multi-Platform Cluster Attack — The Offshore Construction Vessel Constraint

5.5.1 Initiating Event

Coordinated attacks against three or more major Gulf of Mexico deepwater platforms or pipeline hubs, executed within a 72-hour window. Each attack employs the Scenario A or Scenario B template independently; the compound effect is the attack's defining characteristic. Three simultaneous major GoM production outages representing 200,000 to 400,000 barrels per day each — a total of 600,000 to 1,200,000 barrels per day — exceed the Gulf of Mexico basin's recovery capacity at every level simultaneously.

5.5.2 The Offshore Construction Vessel Constraint

The offshore equivalent of the specialty contractor constraint identified in the FC1 cluster attack scenario is the offshore construction vessel fleet. Deep-water pipe-lay vessels, heavy-lift crane vessels, and well intervention vessels are global assets with long mobilization times. There are approximately 8 to 12 vessels worldwide capable of performing the deepwater pipeline work required to replace a PLEM or repair a subsea pipeline at depths greater than 3,000 feet. In normal operations, these vessels are under long-term contract and positioned globally; mobilizing one to the Gulf of Mexico from a North Sea or Brazilian assignment requires weeks.

Three simultaneous GoM deepwater infrastructure events requiring pipe-lay vessel support exceed the available Gulf of Mexico vessel capacity by a factor of two or three. Each platform queues for vessel availability. The adversary who designs the three-platform attack understands this dynamic: the third platform's recovery timeline is not determined by the damage to that platform alone but by the vessel schedule created by the first and second platform recoveries. Recovery timelines that would be 4 to 9 months for a single-platform event extend to 12 to 24 months for the last facility in the queue.

5.5.3 Consequence Envelope: Cluster Attack

Supply impact. 600,000 to 1,200,000 barrels per day of GoM crude production is offline for a sustained period. This represents 3.5 to 7 percent of total U.S. crude production. Strategic Petroleum Reserve drawdown provides a crude supply bridge. Still, it is constrained by SPR drawdown rate limits and the time required for SPR crude to reach the refinery system — 13 to 30 days from authorization to refinery receipt under normal logistics conditions.

Refinery feedstock cascade. Gulf Coast refineries that rely on GoM offshore crude as a significant portion of their feedstock slate will reduce throughput or switch to alternative crude grades. Crude grade switching requires refinery configuration adjustments and may reduce yields of specific products. The compound effect of a GoM cluster attack on refinery throughput and product yields amplifies the supply disruption beyond the production loss alone.

Military logistics impact. The Gulf of Mexico offshore production system supplies crude to the Gulf Coast refinery cluster that produces a significant share of military aviation fuel and diesel. A sustained GoM production outage that reduces Gulf Coast refinery feedstock availability affects military fuel supply with a lag of two to four weeks — the time required for the feedstock reduction to propagate through the refinery system and into the military fuel supply chain.

6. Dual Design Basis Architecture and AI Governance

Consistent with the DBA-OIL Sector Framework (§5.4) and DBA-EN-SF (§4.2), an offshore crude production facility may simultaneously carry a sector-specific physical Design Basis established by this document and an AI Governance Design Basis established under the AI Governance series. The two design bases interact at the DBA level. The Bright Line and Command Broker architecture established in CB-Framework_Cross_Sector v0.2 governs the boundary between the two design bases. The governing foundational instruments are FD-BL (placement), FD-BL-D1 (mode of discharge), and FD-BL-D2 (indication integrity), applied here by reference.

The offshore environment presents a specific challenge for the Bright Line architecture that has no precise onshore analog: communications latency and bandwidth constraints between the platform and the onshore control center. Many deepwater GoM platforms use satellite communications as the primary link to support onshore; in a conflict environment, the adversary

may degrade or spoof satellite communications, creating conditions in which the Command Broker function — which requires a human in the loop for Gen AI advisory outputs — cannot be reliably executed through the normal communications path. The Design Basis requirement for the offshore AI governance interaction is therefore more stringent than its onshore refinery counterpart: the platform must maintain a local Command Broker capability that functions without continuous onshore communications.

Three offshore-specific AI governance interaction points are identified for the FC2 Design Basis:

Production optimization and well management systems. Automated well management systems that continuously adjust choke settings, gas lift rates, and wellhead pressures across multiple wells are ML systems operating below the Bright Line. They are deterministic and formally testable. However, in the offshore context, an adversarially manipulated well management output that drives wellhead conditions outside safe operating bands can initiate a well control event — the highest-consequence failure mode for the offshore facility class. The AI Governance Design Basis must establish that well-managed ML systems are architecturally isolated from the platform IT network, that their outputs are range-checked against independent well integrity monitoring data before execution, and that anomalous well parameter trajectories trigger local platform alerts independent of the onshore communications link.

Emergency shutdown and blowout preventer control systems. ESD and BOP control systems are deterministic ML systems operating below the Bright Line; they must execute without Command Broker interposition when process conditions exceed defined thresholds. Under military-action conditions, the DCS environment in which they operate may be compromised. The Deepwater Horizon progression demonstrated the catastrophic consequences of BOP failure; the DBA-MA extension adds the adversarial manipulation pathway — DCS compromise that inhibits or delays ESD and BOP actuation. The Design Basis requirement is that ESD and BOP control systems are hardwired to independent safety sensors and incapable of being inhibited by DCS software commands, including commands generated by any ML or Gen AI component.

Remotely operated vehicle control and subsea intervention systems. ROV systems used for subsea inspection and intervention are increasingly incorporating ML-assisted navigation, obstacle avoidance, and task execution. These systems operate below the Bright Line when performing deterministic, formally testable navigation functions. However, Gen AI advisory tools that recommend subsea intervention actions — valve closures, connector disconnection, or riser emergency disconnection — are operating above the Bright Line and require Command Broker authorization. In an attack scenario, the adversary may attempt to manipulate ROV systems to prevent or delay the subsea interventions required for emergency response. The Design Basis requires that ROV control systems have a manual override capability that is not dependent on any digital advisory system.

7. Protective Action Thresholds

7.1 Threat Detection and Indicator Taxonomy

7.1.1 Elevated Threat Indicators

Intelligence Community reporting of adversary maritime special operations activity in the Gulf of Mexico or adversary interest in GoM offshore production infrastructure. Anomalous platform control system network activity: unusual access to process historian data, anomalous satellite communications traffic patterns, or unauthorized access attempts to the subsea control system. Unidentified underwater contacts near major platform structures or pipeline hubs, detected by platform sonar or reported by surface vessels. UAS activity in platform airspace is not associated with scheduled supply operations. Requests for GoM pipeline chart data or platform engineering drawings through open or suspicious channels.

7.1.2 Imminent Threat Indicators

Confirmed unauthorized UAS or fixed-wing aircraft in platform exclusion zones. Simultaneous DCS anomalies on two or more platforms within a GoM production cluster. Confirmed subsea contact with platform foundation or pipeline structure. Unexplained production pressure anomalies at a pipeline hub are inconsistent with normal operational variation—confirmed kinetic event at any GoM offshore facility.

7.2 Decision Authority Framework

Platform level: Offshore Installation Manager (OIM) authority to initiate emergency shutdown, well shut-in, crew musters, and evacuation. BSEE and USCG notification for confirmed or suspected well control events, fires, or security incidents. NRC-equivalent notification threshold for events with potential offshore environmental consequences.

Federal coordination: USCG Sector Commander activation for maritime security incidents affecting offshore facilities. BSEE regional office activation for well control or production safety events. DOE ESF-12 activation for supply disruption response. FBI and DOD coordination for events with confirmed or strongly suspected state-actor attribution.

Defense coordination: For events with confirmed or strongly suspected state-actor attribution, National Security Council coordination and Navy involvement in maritime domain awareness and response. The offshore platform attack scenario is inherently a national security event, and the response must engage the interagency national security structure from the outset.

7.3 Federal Coordination Thresholds

Confirmed kinetic strike or subsea sabotage event at any GoM offshore production platform or pipeline hub. Confirmed DCS compromise with suspected nation-state attribution at any offshore production facility. Simultaneous unexplained production outages at two or more major GoM platforms within 72 hours. Hydrocarbon release to the GoM from a cause consistent with

adversary action. Supply disruption affecting military fuel contracts above the defined threshold volumes.

8. Recovery Architecture

8.1 Offshore Recovery Sequencing

Phase 1 — Immediate (0–72 hours). Well shut-in at subsea wellheads. Crew evacuation and personnel accountability. Platform safety assessment by ROV when conditions permit. USCG marine environmental response activation for hydrocarbon release. DCS forensic investigation initiation at the onshore support center. Pipe-lay vessel and well intervention vessel procurement inquiry initiated — contract availability assessment is a Phase 1 action, not Phase 2, because vessel mobilization timelines are the critical path.

Phase 2 — Short-term (72 hours to 8 weeks). ROV damage assessment will be completed when the weather permits. Pipe-lay vessel contracted and mobilized. Subsea repair materials and replacement components identified, and procurement initiated. DCS forensic investigation completed; systems reconstituted before any remote operation of subsea equipment. Alternative crude supply arrangements were activated for refineries dependent on affected platforms—regulatory interaction with BSEE for inspection and restart authorization planning.

Phase 3 — Medium-term (8 weeks to recovery). Subsea pipeline or riser repair by pipe-lay vessel. Topside structural repair where required. Sequential well restart following regulatory clearance. Supply chain monitoring for adversary interdiction of replacement components — particularly specialized subsea equipment with long manufacturing lead times.

8.2 Critical Recovery Constraints

Offshore construction vessel availability. This is the governing constraint for the Scenario A, B, and C consequence envelopes. Deep-water pipe-lay vessels and heavy-lift crane vessels are the scarcest resource in offshore infrastructure recovery, and their global distribution means that mobilization to the GoM from other basins requires weeks. No strategic reserve of offshore construction vessel capacity exists. Pre-established vessel reservation agreements for emergency mobilization — analogous in concept to the electric sector's large power transformer reserve program — would be the most consequential single mitigation for the cluster attack consequence envelope. No such program currently exists.

Subsea component procurement. Deepwater subsea equipment — PLEMs, manifolds, flow line connectors, riser assemblies — is manufactured in limited quantities by a small number of specialized suppliers. Replacement lead times for major subsea components range from 6 to 18 months under normal supply chain conditions. Supply chain interdiction by the adversary targeting the specialized subsea manufacturing base could extend platform recovery timelines

significantly beyond the physical repair timeline. Pre-positioned critical spare components at a GoM onshore staging facility would reduce this constraint.

Regulatory restart authorization. BSEE inspection and restart authorization following a major offshore incident adds weeks to months to the physical repair timeline, as demonstrated by the Deepwater Horizon moratorium precedent. Recovery planning must account for the regulatory timeline as a deterministic component of the recovery path, not a variable to be negotiated after the fact.

Weather access windows. GoM weather, particularly during hurricane season (June through November), constrains offshore construction vessel operations. A recovery that begins in July may lose 8 to 12 weeks of working time to weather constraints before the primary construction season resumes. Attack timing that initiates recovery at the beginning of hurricane season maximizes the effective outage duration through purely environmental means.

8.3 Supply Chain Monitoring

The adversary who executes a Scenario A, B, or C attack understands the recovery critical path as well as the defender does. Supply chain interdiction targeting deep-water pipe-lay vessel contracts, specialized subsea component manufacturers, and DCS hardware suppliers — all of which have concentrated global supply chains — can substantially extend the effective disruption period beyond the physical damage timeline. Recovery planning must include active monitoring of the supply chains for these critical long-lead-time items, with pre-positioned procurement initiated before an attack occurs.

9. Gaps and Recommendations

9.1 Regulatory Gap: Absence of Mandatory Cybersecurity Standards for Offshore Control Systems

Offshore platform control systems — process control networks, subsea control systems, and DCS architectures — have no mandatory cybersecurity standard equivalent to TSA's pipeline security directives or the electric sector's NERC CIP. BSEE's safety case regulations address the physical integrity of safety systems but do not require a cybersecurity assessment of the control system environment. A platform whose DCS is compromised — as in Scenario A — has no regulatory framework requiring the detection capability, architectural isolation, or incident response planning that would limit the cyber component's contribution to the attack consequence. This is the most consequential single gap in the offshore production security framework.

9.2 Offshore Construction Vessel Strategic Reserve

The offshore construction vessel constraint identified in Section 8.2 is the governing recovery gap for the cluster attack scenario. No federal agency has assessed the GoM's recovery capacity

against the Scenario C consequence envelope. DOE, in coordination with BSEE, USCG, and the Bureau of Ocean Energy Management (BOEM), should conduct a GoM offshore infrastructure recovery capacity assessment using the Scenario C consequence envelope as the design basis. The assessment should quantify the vessel capacity gap and evaluate the feasibility of a strategic vessel reservation agreement program analogous to the electric sector's spare transformer reserve.

9.3 Subsea Component Pre-Positioning

The manufacturing lead time for major deepwater subsea components is 6 to 18 months. No federal or industry program requires pre-positioning of critical spare subsea components for emergency recovery. The International Association of Oil and Gas Producers (IOGP) and the Offshore Operators Committee (OOC) have developed mutual-aid frameworks for offshore emergency response equipment; extending these frameworks to cover critical subsea components — beginning with PLEMs and manifolds for the highest-producing GoM fields — would reduce the Scenario A and B recovery timelines by months.

9.4 Pipeline Hub Vulnerability Assessment

The GoM pipeline hub architecture concentrates multi-platform production through a small number of subsea and nearshore gathering points. No systematic security assessment of the GoM pipeline hub vulnerability to subsea sabotage has been publicly reported. BSEE and PHMSA, in coordination with USCG, should commission a classified hub vulnerability assessment using the Scenario B consequence envelope and the Nord Stream operational template as the design basis. The results should inform both physical hardening investment and maritime domain awareness requirements for hub-adjacent sea areas.

9.5 Allied-Nation Coordination

The North Sea and Norwegian Continental Shelf face an analogous maritime threat profile from the same adversary tier. The Scenario B template — subsea pipeline sabotage — was already executed against Nord Stream. The allied offshore production security community has not developed a shared Design Basis framework equivalent to the DBA-OIL series. A NATO energy security working group with access to the DBA-OIL-FC2 analytical framework could provide the foundation for allied offshore infrastructure protection standards and mutual recovery assistance arrangements.

Revision History

Version	Date	Author	Description
0.1	June	T. Roxey	Initial version. Establishes Design Basis Accident — Military Action

	2026		scenarios for the offshore crude production facility class. Issued under DBA-OIL Sector Framework v1.7. Three scenarios were developed: Scenario A (coordinated cyber-kinetic platform attack), Scenario B (subsea pipeline hub sabotage), Scenario C (multi-platform cluster attack — offshore construction vessel constraint). Dual Design Basis architecture established with three offshore-specific AI governance interaction points. Regulatory gap (offshore control system cybersecurity), offshore construction vessel strategic reserve gap, subsea component pre-positioning gap, pipeline hub vulnerability assessment gap, and allied-nation coordination gap were identified.
0.2	June 2026	T. Roxey	DB-layer formalization (Finding 1). §3 narrative DB converted to series-standard event-code tables: EF (EF-1 platform fire/explosion and blowout, EF-2 subsea infrastructure failure, EF-3 export-path failure); NH formalized (NH-1 hurricane, NH-2 loop current/metocean, NH-3 subsea geohazard); OE added (OE-1 allocation/scheduling, OE-2 deferred intervention); XC added (XC-1 electric, XC-2 refining cascade, XC-3 associated-gas). New §4 non-adversarial DBA consequence envelope with SDD: bounding-event identification, SDD metric definition, Tier 1/2/3 taxonomy, SDD acceptance framework, and §4.5 establishing the offshore construction vessel as the governing long-lead constraint (RT_HVLLC analog). v0.1 §3.2 MA bounding event folded into §5.1; §5 DBA-MA extension adds §5.1 foundation and §5.2 four-characteristics table ahead of Scenarios A/B/C (renumbered §§5.3–5.5); v0.1 §3.3 Dual-DB promoted to standalone §6; v0.1 §§5–7 renumbered §§7–9. Parent updated DBA-OIL Sector Framework v1.7 → v1.7; grandparent DBA-EN-SF added at v0.8. Four-filter MA applicability and CB-1–CB-5 noted pending DBA-OIL-MA-SF. Header/footer wiring made explicit. DBA-OIL Sector Framework v1.7. Three scenarios were developed: Scenario A (coordinated cyber-kinetic platform attack), Scenario B (subsea pipeline hub sabotage), Scenario C (multi-platform cluster attack — offshore construction vessel constraint). Dual Design Basis architecture established with three offshore-specific AI governance interaction points. Regulatory gap (offshore control system cybersecurity), offshore construction vessel strategic reserve gap, subsea component pre-positioning gap, pipeline hub vulnerability assessment gap, and allied-nation coordination gap were identified.